

Lagnajeet Panigrahi

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Education

University of California, Berkeley

Berkeley, CA

B.S. Electrical Engineering & Computer Science (EECS) — Expected Graduation: Spring 2028

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Discrete Math, Efficient Algorithms, Machine Learning, Deep Learning, Intro to Quant Finance, Circuits, Signal Processing, Optimization Models

Relevant Experience

EnsureCare.ai

San Francisco, CA

Machine Learning Engineer / Full-Stack Software Engineer

April 2026 – Present

- Built a high-risk patient no-show prediction system with 100K+ patient records, reducing missed appointments by 25%.
- Designed a risk prioritization system using predictive modeling and REST APIs, improving follow-up efficiency by 30%+.
- Orchestrated ETL pipelines across 100k+ healthcare records using PostgreSQL for model training and live risk scoring.
- Developed full-stack features with React, TypeScript, Node.js, and FastAPI, reducing provider review time by 40%+.

Berkeley Artificial Intelligence Research (BAIR)

Berkeley, CA

Machine Learning Engineer & Robotics Researcher, Agile Robotics and Perception Lab

September 2025 – Present

- Designed a multi-head LSTM predictor for long-horizon quadrotor dynamics using 450,000+ data points.
- Studied world models that learn environmental dynamics for prediction, reasoning, and real-time decision-making.
- Investigated latent-space representations to capture motion and improve generalizable robotic behavior.

UMass Chan Medical School Research, Department of Computational Biology

Worcester, MA

Machine Learning Engineer & Bioinformatics Researcher

May 2024 – September 2024

- Developed 6 CNNs for binary classification and regression of gene expression, achieving 70%+ accuracy.
- Conducted statistical analysis of genes and DNA sequences; optimized models through data augmentation.
- Integrated a generative AI model (OpenAI API) into transcription-factor research, reducing data analysis time by 20%.

Projects

Claimd: AI-Assisted Disability Claims Platform [\[GitHub\]](#)

October 2025 – April 2026

CalHacks 12.0 / 2nd Place, Promise Track

- Built AI-powered caseworker portal reducing SSDI review time from 7 months to 2 weeks for 940K applicants.
- Architected serverless dual-agent system on AWS Lambda and SQS with <5s processing and 1000 req/s capacity.
- Built scalable data pipeline processing 390K+ claims with feature engineering and K-means complexity scoring.
- Implemented multi-agent pipeline with Claude Opus 4.6 achieving adherence to SSA's 5-step legal evaluation.
- Developed Transformer-based routing policies for optimal case assignment in hierarchical reviewer workflows.

VeriLaw: Verifiable AI Legal Compliance Platform [\[GitHub\]](#)

October 2025 – November 2025

a16z Hackathon Winner (\$2,000), 1st in Monetization Track, 3rd in Eigenlayer Track

- Built an AI compliance app running LLM document checks inside an EigenCloud Trusted Execution Environment.
- Implemented Flask APIs for secure upload, TEE orchestration, and cryptographic proofs stored in Supabase/PostgreSQL.
- Built a React/TypeScript frontend and Dockerized backend/TEE services for local development and cloud deployment.

Automating Brain Tumor Classification Using CNNs [\[GitHub\]](#)

August 2022 – June 2023

Research presented at MIT, Columbia, and WPI science fairs

- Built a CNN in TensorFlow/Keras to classify brain MRI scans for tumor detection with 90%+ test accuracy.
- Applied image preprocessing and data augmentation to expand the MRI dataset and improve validation accuracy.
- Tuned architecture and hyperparameters to reduce training time per epoch by ~30% while maintaining accuracy.

Leadership

NeuroTech@Berkeley, Software Division Neuromorphic Computing Project Lead

January 2026 – Present

AdaptiveSpikformer: Confidence-Aware Adaptive Computation in Spiking Transformers

- Improved spiking transformer efficiency with SEENN-I early exit, reducing inference compute by 29–39% without loss.

Massachusetts State House of Representatives

June 2024 – August 2024

Technology Legislation Intern for Rep. Hannah Kane

- Researched and presented 10+ bills concerning technology ethics, artificial intelligence, and public policy.

Skills

Languages: Python, Java, C++, JavaScript, SQL

Frameworks: PyTorch, TensorFlow, Scikit-learn, React, FastAPI, Django, PostgreSQL, MongoDB

AI/ML: Transformers, CNNs, LLMs, RAG, Hugging Face, Computer Vision, NLP, Feature Engineering

Tools: Git, Docker, Linux, AWS, GCP, REST APIs, ROS, ROS 2

Awards

a16z 10 Squared Hackathon Winner; 2nd Place, CalHacks 12.0 Promise Track; International Young Researcher's Conference Invitee at Columbia University; Massachusetts State Science Fair 2nd Place; National Chemistry Olympiad Qualified; FIRST Robotics New England District Runner Up; Shrewsbury Senior Scholar